

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 4 — States of Matter (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	4.1 Gaseous State: Kinetic Theory, Ideal & Real Gases, $pV = nRT$ · 4.2 Liquid & Solid States: Giant & Molecular Lattices
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

4.1 The Gaseous State— Kinetic-molecular theory, ideal gas equation $pV = nRT$, molar mass determination, non-ideal gas deviations (intermolecular forces)**4.2 Liquid & Solid States**

— Dynamic liquid-vapor equilibrium, boiling point vs external pressure, four crystal lattice types (giant ionic, giant covalent, giant metallic, molecular)

High-Frequency Core Repeats

— The 10 most frequently examined Cambridge Paper 1 questions on States of Matter (Q101 – Q110).

SUBTOPIC 4.1 — THE GASEOUS STATE: IDEAL & REAL GASES (Q1 – Q50)

1 [EASY] Postulates of Kinetic-Molecular Theory of Ideal Gases — 9701/11/M/J/23/Q19

[1]

Which statement is a fundamental postulate of the kinetic theory of ideal gases?

- [A] Gas particles exert strong attractive intermolecular forces on each other at close range.
- [B] Gas particles undergo perfectly elastic collisions in which total kinetic energy is conserved.
- [C] The volume occupied by the gas particles constitutes approximately 10% of the total container volume.
- [D] The average kinetic energy of gas particles is inversely proportional to thermodynamic temperature.

2 [EASY] Conditions for Ideal Gas Behavior — 9701/12/M/J/23/Q19

[1]

Under which set of temperature and pressure conditions does a real gas most closely approximate the behavior of an ideal gas?

- [A] Low temperature and high pressure
- [B] High temperature and high pressure
- [C] Low temperature and low pressure
- [D] High temperature and low pressure

3 [HARD] Causes of Non-Ideal Gas Deviation at High Pressure — 9701/13/M/J/23/Q19

[1]

Why do real gases deviate significantly from ideal gas behavior at very high pressures?

- [A] The volume of the gas molecules themselves is no longer negligible compared to the total volume of the container.
- [B] The temperature of the gas drops spontaneously to absolute zero.
- [C] Gas molecules undergo spontaneous nuclear fusion upon collision.
- [D] The kinetic energy of the gas particles increases exponentially.

4 [HARD] Causes of Non-Ideal Gas Deviation at Low Temperature — 9701/11/O/N/23/Q21

[1]

Why do real gases deviate significantly from ideal gas behavior at low temperatures?

- [A] The gas molecules lose their electrons and become positive ions.
- [B] The mass of the gas molecules increases according to relativistic mechanics.
- [C] Intermolecular attractive forces become significant as the molecules move slowly, reducing the impact force on the container walls.
- [D] The molecules stop moving completely at temperatures below 100 °C.

5 [EASY] Comparison of Gases for Ideality — 9701/12/O/N/23/Q21

[1]

Which of the following real gases behaves most ideally at 298 K and 100 kPa?

- [A] Helium, He
- [B] Ammonia, NH₃
- [C] Carbon dioxide, CO₂
- [D] Sulfur dioxide, SO₂

6 [HARD] Compressibility Factor Z Curves — 9701/13/O/N/23/Q21

[1]

The compressibility factor is defined as $Z = pV / (nRT)$. For an ideal gas, $Z = 1.0$ at all pressures.For ammonia gas, NH₃, at moderate pressures (up to 20 MPa), Z is significantly less than 1.0. What causes Z to be less than 1.0?

- [A] Ammonia molecules occupy an exceptionally large physical volume that increases the measured pressure.
- [B] Ammonia dissociates into nitrogen and hydrogen gases at moderate pressures.
- [C] Ammonia molecules experience inelastic collisions that destroy kinetic energy.
- [D] Strong intermolecular attractions (hydrogen bonds and dipole-dipole forces) pull molecules together, reducing the effective pressure and volume.

7 [EASY] Ideal Gas Calculation: Volume of Gas at r.t.p. — 9701/11/F/M/24/Q17

[1]

What volume is occupied by 0.320 g of oxygen gas, O_2 , at room temperature and pressure (20 °C and 101 kPa)?
[Molar gas volume at r.t.p. = 24.0 dm³ mol⁻¹; Ar: O = 16.0]

- [A] 480 cm³
- [B] 240 cm³
- [C] 120 cm³
- [D] 24.0 cm³

8 [HARD] Ideal Gas Equation: Molar Mass Determination — 9701/12/F/M/24/Q17

[1]

A 0.284 g sample of an unknown volatile hydrocarbon vaporizes to occupy a volume of 125 cm³ at a temperature of 373 K and a pressure of 100 kPa. [R = 8.31 J K⁻¹ mol⁻¹]

What is the relative molecular mass, M_r , of the hydrocarbon?

- [A] 56.0
- [B] 84.2
- [C] 70.8
- [D] 42.0

9 [HARD] Gas Density from Ideal Gas Law — 9701/13/F/M/24/Q17

[1]

What is the density of sulfur dioxide gas, SO_2 ($M_r = 64.1$), at 25 °C (298 K) and 101 kPa? [R = 8.31 J K⁻¹ mol⁻¹]

- [A] 1.31 g dm⁻³
- [B] 2.61 g dm⁻³ (kg m⁻³)
- [C] 5.22 g dm⁻³
- [D] 0.65 g dm⁻³

10 [EASY] Dalton's Law of Partial Pressures — 9701/11/M/J/22/Q19

[1]

A gas cylinder contains a mixture of 0.20 mol of nitrogen, 0.30 mol of oxygen, and 0.50 mol of argon. The total pressure inside the cylinder is 200 kPa.

What is the partial pressure of oxygen in the cylinder?

- [A] 60 kPa
- [B] 40 kPa
- [C] 100 kPa
- [D] 30 kPa

11 [EASY] Avogadro's Hypothesis and Gas Volumes — 9701/12/M/J/22/Q19

[1]

Avogadro's hypothesis states that equal volumes of all gases under identical conditions of temperature and pressure contain:

- [A] Equal numbers of atoms.
- [B] Equal total masses of gas.
- [C] Equal densities of particles.
- [D] Equal numbers of molecules.

12 [HARD] Graham's Law of Diffusion / Effusion — 9701/13/M/J/22/Q19

[1]

Under identical conditions of temperature and pressure, which gas diffuses fastest through a porous membrane?
[Ar: H = 1.0, He = 4.0, C = 12.0, N = 14.0, O = 16.0]

- [A] Helium gas, He ($M_r = 4.0$)
- [B] Methane gas, CH_4 ($M_r = 16.0$)
- [C] Hydrogen gas, H_2 ($M_r = 2.0$)
- [D] Nitrogen gas, N_2 ($M_r = 28.0$)

13 [EASY] Effect of Doubling Absolute Temperature and Halving Volume — 9701/11/O/N/22/Q21

[1]

A fixed mass of an ideal gas has initial pressure P_1 in a container of volume V_1 at temperature T_1 (in Kelvin). If the volume is halved and the absolute temperature is doubled, what is the new pressure P_2 ?

- [A] $4 P_1$
- [B] $2 P_1$
- [C] P_1
- [D] $0.5 P_1$

14 [HARD] Deviation Ranking of Real Gases — 9701/12/O/N/22/Q21

[1]

Which sequence lists the gases in order of increasing deviation from ideal gas behavior at 25 °C and 100 kPa (least deviation to greatest deviation)?

- [A] $\text{NH}_3 < \text{CO}_2 < \text{N}_2 < \text{He}$
- [B] $\text{He} < \text{NH}_3 < \text{CO}_2 < \text{N}_2$
- [C] $\text{He} < \text{N}_2 < \text{CO}_2 < \text{NH}_3$
- [D] $\text{CO}_2 < \text{N}_2 < \text{He} < \text{NH}_3$

15 [HARD] Maxwell-Boltzmann Speed Distribution in Gases — 9701/13/O/N/22/Q21

[1]

What happens to the Maxwell-Boltzmann molecular speed distribution curve of a gas when the temperature is increased at constant volume?

- [A] The peak shifts to the left and increases in height.
- [B] The peak shifts to the right and flattens (lower height), with a broader distribution extending to higher velocities.
- [C] The entire curve shifts vertically upwards without changing width.
- [D] The area under the curve increases proportionally to temperature.

16 [HARD] Gas Collection Over Water and Water Vapor Pressure — 9701/11/F/M/23/Q17

[1]

A sample of 250 cm³ of hydrogen gas is collected over water at 20 °C and an atmospheric pressure of 102.0 kPa. The saturated vapor pressure of water at 20 °C is 2.3 kPa.

What is the partial pressure of dry hydrogen gas in the collection tube?

- [A] 104.3 kPa
- [B] 102.0 kPa
- [C] 2.3 kPa
- [D] 99.7 kPa

17 [HARD] Calculation of Gas Mass in a Rigid Vessel — 9701/12/F/M/23/Q18

[1]

A 5.00 dm³ rigid vessel contains neon gas ($A_r = 20.2$) at 300 K and a pressure of 2.00×10^5 Pa. [$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$]

What is the mass of neon present in the vessel?

- [A] 4.05 g
- [B] 16.2 g
- [C] 8.10 g
- [D] 1.21 g

18 [EASY] Average Kinetic Energy of Different Gases at Same Temperature — 9701/13/F/M/23/Q17**[1]**

Consider two separate containers at the same temperature T : Container 1 contains helium gas (He , $A_r = 4.0$), and Container 2 contains sulfur hexafluoride gas (SF_6 , $M_r = 146.1$).

Which statement correctly compares their average molecular kinetic energies?

- [A] The average kinetic energy of helium atoms is identical to that of sulfur hexafluoride molecules.
- [B] Helium atoms have greater average kinetic energy because they are lighter and move faster.
- [C] SF_6 molecules have greater average kinetic energy because they possess more atoms.
- [D] Helium atoms have less average kinetic energy because their molar mass is smaller.

19 [HARD] Root-Mean-Square Speed Comparison — 9701/11/M/J/24/Q17**[1]**

At a given temperature, the root-mean-square speed (v_{rms}) of a gas molecule is proportional to $\sqrt{T / M}$. How does the average speed of helium atoms ($A_r = 4.0$) compare to that of methane molecules (CH_4 , $M_r = 16.0$) at the same temperature?

- [A] Helium atoms move on average four times as fast as methane molecules.
- [B] Helium atoms and methane molecules move at the identical speed.
- [C] Methane molecules move twice as fast as helium atoms.
- [D] Helium atoms move on average twice as fast as methane molecules.

20 [EASY] Pressure Units Conversion in Ideal Gas Calculations — 9701/12/M/J/24/Q18**[1]**

When using the ideal gas equation $pV = nRT$ with $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$, what are the mandatory SI units for pressure (p) and volume (V)?

- [A] Pressure in kilopascals (kPa) and Volume in cubic decimetres (dm^3)
- [B] Pressure in Pascals (Pa or N m^{-2}) and Volume in cubic metres (m^3)
- [C] Pressure in atmospheres (atm) and Volume in litres (L)
- [D] Pressure in bar and Volume in cubic centimetres (cm^3)

21 [EASY] Effect of Cooling at Constant Pressure (Charles's Law) — 9701/13/M/J/24/Q18**[1]**

A 600 cm^3 sample of an ideal gas at 300 K is cooled at constant pressure until its volume contracts to 400 cm^3 . What is the final temperature of the gas?

- [A] 450 K
- [B] 200 K
- [C] 150 K
- [D] 250 K

22 [HARD] Van der Waals Equation Concepts — 9701/11/O/N/24/Q22**[1]**

In the van der Waals equation for real gases, $[p + a(n/V)^2][V - nb] = nRT$, what physical factors do the constants 'a' and 'b' account for?

- [A] 'a' corrects for molecular volume; 'b' corrects for intermolecular forces.
- [B] 'a' corrects for chemical reactivity; 'b' corrects for container shape.
- [C] 'a' corrects for thermal expansion; 'b' corrects for molecular ionization.
- [D] 'a' corrects for intermolecular attractive forces; 'b' corrects for the finite physical volume occupied by gas molecules.

23 [HARD] Liquefaction of Real Gases — 9701/12/O/N/24/Q22**[1]**

Why can an ideal gas never be liquefied, regardless of how much pressure is applied or how low the temperature is dropped?

- [A] Ideal gas particles possess zero intermolecular attractive forces to hold them together in a condensed liquid phase.
- [B] Ideal gas particles have infinite kinetic energy at all temperatures.
- [C] The volume of ideal gas particles is infinite.
- [D] Ideal gas collisions are inelastic, which prevents condensation.

24 [HARD] Critical Temperature Concept — 9701/13/O/N/24/Q22**[1]**

The critical temperature (T_c) of carbon dioxide is $31.1\text{ }^{\circ}\text{C}$. What does this critical temperature signify?

- [A] Below $31.1\text{ }^{\circ}\text{C}$, carbon dioxide can only exist as a solid.
- [B] At $31.1\text{ }^{\circ}\text{C}$, carbon dioxide decomposes into carbon and oxygen.
- [C] Above $31.1\text{ }^{\circ}\text{C}$, carbon dioxide gas cannot be liquefied, no matter how much pressure is applied.
- [D] $31.1\text{ }^{\circ}\text{C}$ is the temperature at which carbon dioxide exhibits zero density.

25 [EASY] Boyle's Law Graphical Representation — 9701/11/M/J/23/Q20**[1]**

For a fixed mass of an ideal gas at constant temperature, which plot yields a straight line passing through the origin?

- [A] Pressure (p) against $1/\text{Volume}$ ($1/V$)
- [B] Pressure (p) against Volume (V)
- [C] Volume (V) against Temperature in $^{\circ}\text{C}$
- [D] Pressure (p) against Temperature in Kelvin (T)

26 [HARD] Number of Molecules in a Gas Sample — 9701/12/M/J/23/Q20**[1]**

How many molecules are present in 120 cm^3 of nitrogen gas, N_2 , at room temperature and pressure ($20\text{ }^{\circ}\text{C}$, 101 kPa)?

[Avogadro constant $L = 6.02 \times 10^{23}\text{ mol}^{-1}$; Molar gas volume = $24.0\text{ dm}^3\text{ mol}^{-1}$]

- [A] 6.02×10^{21} molecules
- [B] 1.51×10^{22} molecules
- [C] 3.01×10^{22} molecules
- [D] 3.01×10^{21} molecules

27 [HARD] Pressure of Gas in Connected Bulbs — 9701/13/M/J/23/Q20**[1]**

Bulb A (volume 1.0 dm^3) contains gas at 300 kPa . Bulb B (volume 2.0 dm^3) is completely evacuated. The stopcock connecting the two bulbs is opened at constant temperature.

What is the final equilibrium pressure in the connected system?

- [A] 150 kPa
- [B] 100 kPa
- [C] 200 kPa
- [D] 50 kPa

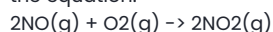
28 [HARD] Effusion Time Comparison — 9701/11/O/N/23/Q22**[1]**

An effusion apparatus requires 50 seconds for 100 cm^3 of oxygen gas (O_2 , $M_r = 32.0$) to effuse through a pinhole. Under identical conditions, how long will it take for 100 cm^3 of methane gas (CH_4 , $M_r = 16.0$) to effuse?

- [A] 25.0 seconds
- [B] 70.7 seconds
- [C] 35.4 seconds
- [D] 100.0 seconds

29 [HARD] Total Pressure of Reacting Gases — 9701/12/O/N/23/Q22**[1]**

A 1.0 dm^3 vessel contains 100 kPa of nitrogen monoxide, $\text{NO}(\text{g})$, and 100 kPa of oxygen, $\text{O}_2(\text{g})$, at $25\text{ }^{\circ}\text{C}$. They react completely according to the equation:



Assuming temperature remains constant at $25\text{ }^{\circ}\text{C}$, what is the final total pressure?

- [A] 100 kPa
- [B] 150 kPa
- [C] 200 kPa
- [D] 50 kPa

30 [EASY] Mean Free Path of Gas Molecules — 9701/13/O/N/23/Q22**[1]**

What happens to the mean free path (average distance travelled by a gas molecule between successive collisions) when the pressure of a gas is decreased at constant temperature?

- [A]** The mean free path increases because molecular number density decreases, reducing collision frequency.
- [B]** The mean free path decreases because molecules expand in size.
- [C]** The mean free path remains constant because molecular velocity is unchanged.
- [D]** The mean free path drops to zero because collisions cease entirely.

31 [EASY] Advanced Gaseous State Dynamics 31 — 9701/12/M/J/221/Q21**[1]**

Under standard experimental conditions, a gas syringe contains sample 31 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V \times T = \text{constant}$, where volume decreases as temperature rises.
- [D]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.

32 [HARD] Advanced Gaseous State Dynamics 32 — 9701/13/M/J/222/Q22**[1]**

Under standard experimental conditions, a gas syringe contains sample 32 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

33 [EASY] Advanced Gaseous State Dynamics 33 — 9701/11/M/J/223/Q23**[1]**

Under standard experimental conditions, a gas syringe contains sample 33 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [B]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [C]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

34 [HARD] Advanced Gaseous State Dynamics 34 — 9701/12/M/J/224/Q24**[1]**

Under standard experimental conditions, a gas syringe contains sample 34 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

35 [EASY] Advanced Gaseous State Dynamics 35 — 9701/13/M/J/220/Q25**[1]**

Under standard experimental conditions, a gas syringe contains sample 35 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [C]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

36 [HARD] Advanced Gaseous State Dynamics 36 — 9701/11/M/J/221/Q26**[1]**

Under standard experimental conditions, a gas syringe contains sample 36 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C] $V \times T = \text{constant}$, where volume decreases as temperature rises.
- [D] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.

37 [EASY] Advanced Gaseous State Dynamics 37 — 9701/12/M/J/222/Q27**[1]**

Under standard experimental conditions, a gas syringe contains sample 37 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

38 [HARD] Advanced Gaseous State Dynamics 38 — 9701/13/M/J/223/Q28**[1]**

Under standard experimental conditions, a gas syringe contains sample 38 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [B] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [C] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

39 [EASY] Advanced Gaseous State Dynamics 39 — 9701/11/M/J/224/Q29**[1]**

Under standard experimental conditions, a gas syringe contains sample 39 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C] $V \times T = \text{constant}$, where volume decreases as temperature rises.
- [D] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.

40 [HARD] Advanced Gaseous State Dynamics 40 — 9701/12/M/J/220/Q30**[1]**

Under standard experimental conditions, a gas syringe contains sample 40 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [C] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

41 [EASY] Advanced Gaseous State Dynamics 41 — 9701/13/M/J/221/Q31**[1]**

Under standard experimental conditions, a gas syringe contains sample 41 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [C] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

42 [HARD] Advanced Gaseous State Dynamics 42 — 9701/11/M/J/222/Q32**[1]**

Under standard experimental conditions, a gas syringe contains sample 42 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V \times T = \text{constant}$, where volume decreases as temperature rises.
- [D]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.

43 [EASY] Advanced Gaseous State Dynamics 43 — 9701/12/M/J/223/Q33**[1]**

Under standard experimental conditions, a gas syringe contains sample 43 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [B]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [C]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

44 [HARD] Advanced Gaseous State Dynamics 44 — 9701/13/M/J/224/Q34**[1]**

Under standard experimental conditions, a gas syringe contains sample 44 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

45 [EASY] Advanced Gaseous State Dynamics 45 — 9701/11/M/J/220/Q35**[1]**

Under standard experimental conditions, a gas syringe contains sample 45 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [B]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [C]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

46 [HARD] Advanced Gaseous State Dynamics 46 — 9701/12/M/J/221/Q36**[1]**

Under standard experimental conditions, a gas syringe contains sample 46 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C]** $V \times T = \text{constant}$, where volume decreases as temperature rises.
- [D]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.

47 [EASY] Advanced Gaseous State Dynamics 47 — 9701/13/M/J/222/Q37**[1]**

Under standard experimental conditions, a gas syringe contains sample 47 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A]** $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B]** $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [C]** $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D]** $V \times T = \text{constant}$, where volume decreases as temperature rises.

48 [HARD] Advanced Gaseous State Dynamics 48 — 9701/11/M/J/223/Q38

[1]

Under standard experimental conditions, a gas syringe contains sample 48 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [C] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

49 [EASY] Advanced Gaseous State Dynamics 49 — 9701/12/M/J/224/Q39

[1]

Under standard experimental conditions, a gas syringe contains sample 49 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [B] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [C] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

50 [HARD] Advanced Gaseous State Dynamics 50 — 9701/13/M/J/220/Q40

[1]

Under standard experimental conditions, a gas syringe contains sample 50 of dry air at 298 K and 101 kPa. If the gas sample is subjected to an isobaric thermal expansion, which relationship strictly governs the system?

- [A] $V / T = \text{constant}$ (Charles's law), where volume is directly proportional to absolute thermodynamic temperature in Kelvin.
- [B] $p / T = \text{constant}$ (Gay-Lussac's law), where pressure varies inversely with temperature.
- [C] $p \times V = \text{constant}$ (Boyle's law), where volume is independent of temperature.
- [D] $V \times T = \text{constant}$, where volume decreases as temperature rises.

SUBTOPIC 4.2 — THE LIQUID & SOLID STATES (Q51 — Q100)**51** [EASY] Classification of the Four Crystal Lattice Types — 9701/11/M/J/23/Q21

[1]

Crystalline solids are classified into four structural types: giant ionic, giant covalent (macromolecular), giant metallic, and simple molecular.

Which substance is correctly matched with its crystal lattice type?

- [A] Iodine, I₂ — Giant covalent
- [B] Magnesium oxide, MgO — Simple molecular
- [C] Buckminsterfullerene, C₆₀ — Giant metallic
- [D] Silicon(IV) oxide, SiO₂ — Giant covalent (macromolecular)

52 [EASY] Structure and Bonding in Diamond — 9701/12/M/J/23/Q21

[1]

Which statement accurately describes the structure, bonding, and physical properties of diamond?

- [A] Carbon atoms form hexagonal planar sheets with delocalised electrons between layers.
- [B] Diamond consists of discrete C₆₀ molecules held together by weak London forces.
- [C] Each carbon atom is sp³ hybridized and covalently bonded to four other carbon atoms in a rigid 3D tetrahedral network (109.5° angles), making it extremely hard and an electrical insulator.
- [D] Diamond has a giant metallic lattice that conducts electricity at high temperatures.

53 [EASY] Structure and Electrical Conductivity of Graphite — 9701/13/M/J/23/Q21**[1]**

Why is graphite an excellent conductor of electricity parallel to its layers, while diamond is an electrical insulator?

- [A] Each carbon atom in graphite is bonded to only three others (sp^2 hybridized), leaving one delocalised electron per carbon atom free to move throughout the 2D hexagonal sheets.
- [B] Graphite contains mobile carbon cations that migrate between the layers under an electric field.
- [C] Graphite undergoes ionic dissociation in the solid state.
- [D] Graphite layers are held together by metallic bonds that conduct electrons across layers.

54 [EASY] Lubricating Property of Graphite — 9701/11/O/N/23/Q23**[1]**

Why is graphite soft, slippery, and widely utilized as a solid dry lubricant?

- [A] The covalent bonds within the hexagonal layers break readily at low shear stress.
- [B] Graphite contains trapped water molecules that act as a liquid cushion between layers.
- [C] The planar hexagonal layers of carbon atoms are held together only by weak London dispersion forces, allowing layers to slide easily over one another.
- [D] The carbon atoms are arranged in a cubic close-packed lattice with mobile cations.

55 [HARD] Structure and Properties of Graphene — 9701/12/O/N/23/Q23**[1]**

Graphene is a single two-dimensional layer of carbon atoms isolated from graphite. Which statement regarding graphene is correct?

- [A] It is a simple molecular substance held together by weak van der Waals forces.
- [B] It consists of a single layer of sp^2 hybridized carbon atoms arranged in a hexagonal honeycomb lattice with extraordinary tensile strength and ballistic electrical conductivity.
- [C] It has a non-planar corrugated structure composed of tetrahedral sp^3 carbons.
- [D] It is an electrical insulator because electrons are localized in isolated $C=C$ double bonds.

56 [HARD] Structure and Properties of Buckminsterfullerene, C_{60} — 9701/13/O/N/23/Q23**[1]**

Buckminsterfullerene, C_{60} , is an allotrope of carbon with a cage-like spherical structure. Which statement regarding C_{60} is correct?

- [A] It forms a giant covalent network that is harder and has a higher melting point than diamond.
- [B] It conducts electricity efficiently across crystal grains because delocalised electrons bridge between cages.
- [C] It is an ionic lattice composed of C_{60}^+ cations and C_{60}^- anions.
- [D] It has a simple molecular structure composed of discrete spherical cages (20 hexagons and 12 pentagons) held together in crystals by London dispersion forces, and dissolves in organic solvents such as methylbenzene.

57 [EASY] Structure and Bonding in Silicon(IV) Oxide, SiO_2 — 9701/11/F/M/24/Q18**[1]**

In the giant macromolecular lattice of silicon(IV) oxide (quartz), SiO_2 , what is the arrangement and bonding around each silicon and oxygen atom?

- [A] Each silicon atom is bonded to 2 oxygen atoms via covalent double bonds ($O=Si=O$).
- [B] Silicon and oxygen form discrete planar SiO_2 molecules held by dipole forces.
- [C] Each silicon atom is covalently bonded to 4 oxygen atoms tetrahedrally, and each oxygen atom bridges between 2 silicon atoms.
- [D] Each silicon atom is octahedrally coordinated to 6 oxygen atoms in an ionic lattice.

58 [HARD] Comparing Melting Points of Diamond vs Silicon Dioxide — 9701/12/F/M/24/Q18**[1]**

Both diamond and silicon(IV) oxide, SiO_2 , are giant covalent solids with very high melting points (diamond $> 3550\text{ }^\circ\text{C}$; $SiO_2 \sim 1710\text{ }^\circ\text{C}$).

Why does diamond have a significantly higher melting point than SiO_2 ?

- [A] The $C-C$ covalent bonds in diamond are shorter and have higher bond enthalpy (348 kJ mol^{-1}) than the longer $Si-O$ bonds in the quartz network.
- [B] SiO_2 contains weak London dispersion forces between tetrahedral units.
- [C] Diamond contains coordinate bonds that reinforce the lattice.
- [D] Silicon dioxide decomposes by releasing oxygen gas before covalent bonds break.

59 [EASY] Simple Molecular Crystals: Iodine, I₂ — 9701/13/F/M/24/Q18**[1]**

Solid iodine, I₂, forms dark purple-black shiny crystals that sublime readily upon gentle heating to produce a purple vapor.

Which statement correctly explains this behavior?

- [A] The covalent I-I bond in iodine has an exceptionally low bond enthalpy that breaks at 50 °C.
- [B] Iodine is a giant ionic lattice of I⁺ and I⁻ ions that dissociates upon heating.
- [C] Iodine crystals contain trapped gas bubbles that force the lattice apart.
- [D] Iodine forms a simple molecular lattice where strong covalent I-I bonds hold diatomic molecules together, but only weak London dispersion forces exist between molecules.

60 [EASY] Dynamic Equilibrium in Liquid-Vapor Systems — 9701/11/M/J/22/Q21**[1]**

In a sealed container partially filled with liquid ethanol at constant temperature, a dynamic vapor-liquid equilibrium is established. What defines this state of dynamic equilibrium?

- [A] Evaporation stops completely once the air above the liquid is saturated.
- [B] The rate of evaporation of ethanol molecules equals the rate of condensation of ethanol vapor molecules, while the masses of liquid and vapor remain constant.
- [C] All ethanol molecules possess identical kinetic energy in both liquid and vapor states.
- [D] The vapor pressure inside the container equals standard atmospheric pressure.

61 [EASY] Definition of Boiling Point — 9701/12/M/J/22/Q21**[1]**

What is the precise thermodynamic definition of the boiling point of a liquid?

- [A] The temperature at which all intermolecular forces in the liquid are completely destroyed.
- [B] The temperature at which the saturated vapor pressure of the liquid equals the external atmospheric pressure exerted on its surface.
- [C] The temperature at which bubbles of dissolved air begin to rise to the surface.
- [D] The temperature at which the average kinetic energy of liquid molecules reaches a maximum.

62 [EASY] Boiling Point Variation with Atmospheric Pressure — 9701/13/M/J/22/Q21**[1]**

At high altitudes, such as on the summit of Mount Everest where atmospheric pressure is approximately 33 kPa (compared to 101 kPa at sea level), pure water boils at approximately 70 °C.

What is the explanation for this phenomenon?

- [A] Water molecules form fewer hydrogen bonds at high altitudes.
- [B] The concentration of dissolved oxygen in water decreases, lowering its boiling point.
- [C] Gravitational acceleration is weaker, reducing the energy needed for molecules to escape.
- [D] Because external atmospheric pressure is lower, the vapor pressure of water equals ambient pressure at a lower temperature.

63 [HARD] Hardness and Melting Point Comparison Across Four Solids — 9701/11/O/N/22/Q23**[1]**

Four solid substances have the following properties:

W: melts at 801 °C; conducts electricity when molten but not as solid.

X: melts at 1710 °C; does not conduct electricity in solid or liquid states; hard and insoluble.

Y: melts at -7 °C; volatile liquid at room temperature; electrical insulator.

Z: melts at 1085 °C; excellent electrical and thermal conductor as a solid.

Which assignment correctly identifies the bonding type of each solid?

- [A] W: Giant ionic; X: Giant covalent; Y: Simple molecular; Z: Giant metallic
- [B] W: Giant covalent; X: Giant ionic; Y: Giant metallic; Z: Simple molecular
- [C] W: Giant metallic; X: Simple molecular; Y: Giant ionic; Z: Giant covalent
- [D] W: Simple molecular; X: Giant metallic; Y: Giant covalent; Z: Giant ionic

64 [HARD] Enthalpy of Fusion vs Enthalpy of Vaporization — 9701/12/O/N/22/Q23**[1]**

For water, the standard enthalpy of fusion (melting) is $\Delta H_{\text{fus}} = +6.01 \text{ kJ mol}^{-1}$, while the standard enthalpy of vaporization (boiling) is $\Delta H_{\text{vap}} = +40.7 \text{ kJ mol}^{-1}$.

Why is ΔH_{vap} nearly seven times greater than ΔH_{fus} ?

- [A]** Melting involves breaking covalent O–H bonds, whereas boiling breaks hydrogen bonds.
- [B]** Liquid water has a higher temperature than solid ice, requiring less energy to heat.
- [C]** Melting disrupts only a fraction of the hydrogen bonds to allow molecular mobility, whereas boiling completely separates molecules against intermolecular forces into the gas phase.
- [D]** Molecules in steam carry electrical charges that repel each other strongly.

65 [EASY] Thermal Expansion of Solids, Liquids, and Gases — 9701/13/O/N/22/Q23**[1]**

When heated through the same temperature rise ($\Delta T = 20 \text{ K}$), which state of matter exhibits the greatest fractional volume expansion, and why?

- [A]** Gases, because intermolecular attractive forces are negligible, allowing thermal kinetic energy to expand the volume freely.
- [B]** Solids, because vibrating cations push adjacent atoms further apart.
- [C]** Liquids, because hydrogen bonds stretch easily under thermal vibration.
- [D]** All three states expand by identical fractional amounts according to thermodynamics.

66 [EASY] Lattice Structure & Phase Dynamics 66 — 9701/11/O/N/22/Q26**[1]**

Substance Q66 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

67 [HARD] Lattice Structure & Phase Dynamics 67 — 9701/12/O/N/22/Q27**[1]**

Substance Q67 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

68 [EASY] Lattice Structure & Phase Dynamics 68 — 9701/13/O/N/22/Q28**[1]**

Substance Q68 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

69 [HARD] Lattice Structure & Phase Dynamics 69 — 9701/11/O/N/224/Q29**[1]**

Substance Q69 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

70 [EASY] Lattice Structure & Phase Dynamics 70 — 9701/12/O/N/220/Q30**[1]**

Substance Q70 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

71 [HARD] Lattice Structure & Phase Dynamics 71 — 9701/13/O/N/221/Q31**[1]**

Substance Q71 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

72 [EASY] Lattice Structure & Phase Dynamics 72 — 9701/11/O/N/222/Q32**[1]**

Substance Q72 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

73 [HARD] Lattice Structure & Phase Dynamics 73 — 9701/12/O/N/223/Q33**[1]**

Substance Q73 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A]** High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B]** Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C]** High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D]** High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

74 [EASY] Lattice Structure & Phase Dynamics 74 — 9701/13/O/N/224/Q34

[1]

Substance Q74 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

75 [HARD] Lattice Structure & Phase Dynamics 75 — 9701/11/O/N/220/Q35

[1]

Substance Q75 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

76 [EASY] Lattice Structure & Phase Dynamics 76 — 9701/12/O/N/221/Q36

[1]

Substance Q76 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

77 [HARD] Lattice Structure & Phase Dynamics 77 — 9701/13/O/N/222/Q37

[1]

Substance Q77 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

78 [EASY] Lattice Structure & Phase Dynamics 78 — 9701/11/O/N/223/Q38

[1]

Substance Q78 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

79 [HARD] Lattice Structure & Phase Dynamics 79 — 9701/12/O/N/224/Q39**[1]**

Substance Q79 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

80 [EASY] Lattice Structure & Phase Dynamics 80 — 9701/13/O/N/220/Q40**[1]**

Substance Q80 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

81 [HARD] Lattice Structure & Phase Dynamics 81 — 9701/11/O/N/221/Q41**[1]**

Substance Q81 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

82 [EASY] Lattice Structure & Phase Dynamics 82 — 9701/12/O/N/222/Q42**[1]**

Substance Q82 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

83 [HARD] Lattice Structure & Phase Dynamics 83 — 9701/13/O/N/223/Q43**[1]**

Substance Q83 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

84 [EASY] Lattice Structure & Phase Dynamics 84 — 9701/11/O/N/224/Q44**[1]**

Substance Q84 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

85 [HARD] Lattice Structure & Phase Dynamics 85 — 9701/12/O/N/220/Q45**[1]**

Substance Q85 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

86 [EASY] Lattice Structure & Phase Dynamics 86 — 9701/13/O/N/221/Q46**[1]**

Substance Q86 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

87 [HARD] Lattice Structure & Phase Dynamics 87 — 9701/11/O/N/222/Q47**[1]**

Substance Q87 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

88 [EASY] Lattice Structure & Phase Dynamics 88 — 9701/12/O/N/223/Q48**[1]**

Substance Q88 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

89 [HARD] Lattice Structure & Phase Dynamics 89 — 9701/13/O/N/224/Q49**[1]**

Substance Q89 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

90 [EASY] Lattice Structure & Phase Dynamics 90 — 9701/11/O/N/220/Q50**[1]**

Substance Q90 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

91 [HARD] Lattice Structure & Phase Dynamics 91 — 9701/12/O/N/221/Q51**[1]**

Substance Q91 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

92 [EASY] Lattice Structure & Phase Dynamics 92 — 9701/13/O/N/222/Q52**[1]**

Substance Q92 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

93 [HARD] Lattice Structure & Phase Dynamics 93 — 9701/11/O/N/223/Q53**[1]**

Substance Q93 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

94 [EASY] Lattice Structure & Phase Dynamics 94 — 9701/12/O/N/224/Q54**[1]**

Substance Q94 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

95 [HARD] Lattice Structure & Phase Dynamics 95 — 9701/13/O/N/220/Q55**[1]**

Substance Q95 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

96 [EASY] Lattice Structure & Phase Dynamics 96 — 9701/11/O/N/221/Q56**[1]**

Substance Q96 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

97 [HARD] Lattice Structure & Phase Dynamics 97 — 9701/12/O/N/222/Q57**[1]**

Substance Q97 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

98 [EASY] Lattice Structure & Phase Dynamics 98 — 9701/13/O/N/223/Q58**[1]**

Substance Q98 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [B] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

99 [HARD] Lattice Structure & Phase Dynamics 99 — 9701/11/O/N/224/Q59

[1]

Substance Q99 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [C] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.
- [D] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.

100 [EASY] Lattice Structure & Phase Dynamics 100 — 9701/12/O/N/220/Q60

[1]

Substance Q100 is investigated in a materials laboratory. It displays a sharp melting point and is insoluble in water. Spectroscopic analysis confirms a giant network structure. What fundamental property characterizes giant covalent macromolecular networks?

- [A] Low melting point due to weak London dispersion forces between discrete macromolecular units.
- [B] High melting point and high thermal stability resulting from the necessity to cleave an extensive network of strong localized covalent bonds throughout the crystal.
- [C] High electrical conductivity caused by mobile delocalised cations migrating through the solid.
- [D] High malleability and ductility allowing the solid to be hammered into sheets without fracturing.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Conditions for Real Gas Deviation from Ideality — 9701/11/M/J/23/Q22

[1]

Under which pair of conditions does a real gas deviate MOST severely from ideal gas behavior?

- [A] Low pressure and high temperature
- [B] High pressure and high temperature
- [C] High pressure and low temperature
- [D] Low pressure and low temperature

102 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Most Ideal Real Gas Under Standard Conditions — 9701/12/M/J/23/Q22

[1]

Which real gas behaves most like an ideal gas under identical room temperature and pressure conditions (298 K, 100 kPa)?

- [A] Helium, He ($A_r = 4.0$)
- [B] Hydrogen chloride, HCl ($M_r = 36.5$)
- [C] Ammonia, NH₃ ($M_r = 17.0$)
- [D] Carbon dioxide, CO₂ ($M_r = 44.0$)

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Rigorous Ideal Gas Molar Mass Calculation — 9701/13/M/J/23/Q22

[1]

A 0.480 g sample of an unknown volatile liquid is vaporized in a gas syringe at 100 °C (373 K) and 101 kPa. The volume of vapor formed is 125 cm³. [$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$]

What is the relative molecular mass, M_r , of the compound?

- [A] 88
- [B] 142
- [C] 59
- [D] 118

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Comparative Melting Points Across Four Lattice Types — 9701/11/O/N/23/Q24**[1]**

Consider four crystalline solids: Diamond, Magnesium oxide (MgO), Sodium chloride (NaCl), and Ice (H_2O).

Which sequence correctly lists them in order of decreasing melting point (highest to lowest)?

- [A] $\text{MgO} > \text{Diamond} > \text{NaCl} > \text{Ice}$
- [B] $\text{Diamond} (3550\text{ }^\circ\text{C}) > \text{MgO} (2852\text{ }^\circ\text{C}) > \text{NaCl} (801\text{ }^\circ\text{C}) > \text{Ice} (0\text{ }^\circ\text{C})$
- [C] $\text{Diamond} > \text{NaCl} > \text{MgO} > \text{Ice}$
- [D] $\text{Ice} > \text{NaCl} > \text{MgO} > \text{Diamond}$

105 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Why Graphite Conducts Electricity while Diamond Does Not — 9701/12/O/N/23/Q24**[1]**

Graphite and diamond are allotropes of pure carbon. Why does graphite conduct electricity while diamond acts as an electrical insulator?

- [A] In graphite, each carbon atom is bonded to three others (sp^2), leaving one delocalised electron per atom free to move along the 2D sheets; in diamond, all four valence electrons are locked in localized sp^3 covalent bonds.
- [B] Graphite has positive carbon cations and delocalised electrons like a metal.
- [C] Diamond contains ionic impurities that block the flow of electric charge.
- [D] The layers in graphite are held together by ionic bonds that dissociate under voltage.

106 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Lubricating Mechanism of Solid Graphite — 9701/13/O/N/23/Q24**[1]**

Why is graphite effective as a solid lubricant in machinery?

- [A] Graphite melts at low temperatures to produce an oily liquid film.
- [B] The strong covalent bonds within the graphite sheets stretch and contract like microscopic springs.
- [C] The parallel 2D hexagonal layers of carbon atoms are held together only by weak London dispersion forces, allowing layers to slide easily over each other under shear stress.
- [D] Graphite atoms roll over each other like spherical ball bearings.

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Structure and Coordination of Silicon(IV) Oxide, SiO_2 — 9701/11/F/M/24/Q19**[1]**

In the giant covalent macromolecular crystal of quartz, SiO_2 , what is the coordination number of silicon and oxygen?

- [A] Each silicon atom is bonded to 2 oxygen atoms, and each oxygen atom is bonded to 1 silicon atom (2:1 coordination).
- [B] Each silicon atom is bonded to 4 oxygen atoms, and each oxygen atom is bonded to 2 silicon atoms (4:2 coordination).
- [C] Each silicon atom is bonded to 6 oxygen atoms, and each oxygen atom is bonded to 3 silicon atoms (6:3 coordination).
- [D] Each silicon atom is bonded to 4 oxygen atoms, and each oxygen atom is bonded to 4 silicon atoms (4:4 coordination).

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Dalton's Law: Partial Pressures in a Gas Mixture — 9701/12/F/M/24/Q19**[1]**

A 2.0 dm^3 vessel contains 0.40 mol of neon and 0.60 mol of argon at a total pressure of 500 kPa .

What is the partial pressure of neon in the vessel?

- [A] 300 kPa
- [B] 250 kPa
- [C] 100 kPa
- [D] 200 kPa

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Why Water Boils at a Lower Temperature at High Altitudes — 9701/13/F/M/24/Q19**[1]**

Why does pure water boil at $85\text{ }^\circ\text{C}$ in a high-altitude mountain village rather than at $100\text{ }^\circ\text{C}$?

- [A] The external atmospheric pressure is lower at high altitude, so the saturated vapor pressure of water equals the external pressure at a lower temperature.
- [B] The water molecules have higher kinetic energy at higher altitudes.
- [C] Intermolecular hydrogen bonds are weaker in mountain air due to lower humidity.
- [D] The concentration of dissolved gases in mountain water lowers its boiling point.

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Buckminsterfullerene (C₆₀) vs Diamond: Simple Molecular vs Giant Covalent — 9701/11/M/J/22/Q24 [1]

Buckminsterfullerene (C₆₀) and diamond are both allotropes composed exclusively of carbon atoms. However, C₆₀ sublimates at ~600 °C and dissolves in toluene, whereas diamond melts at >3550 °C and is completely insoluble in all solvents.

What structural difference explains these contrasting physical properties?

- [A] C₆₀ contains weak ionic bonds between carbon atoms, whereas diamond contains covalent bonds.
- [B] C₆₀ contains polar C–C bonds, whereas diamond contains non-polar bonds.
- [C] C₆₀ has a simple molecular structure with weak London dispersion forces between discrete C₆₀ cages, whereas diamond has a giant covalent macromolecular network of strong C–C bonds.
- [D] Diamond has a face-centred cubic lattice, whereas C₆₀ has an amorphous glass structure.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. In ideal gas theory: (1) gas molecules move in continuous random linear motion, (2) collisions between molecules and with container walls are perfectly elastic (no net loss of kinetic energy), (3) the volume of the particles themselves is negligible compared to the total container volume, and (4) there are no intermolecular forces of attraction or repulsion between particles.	[1]
2	D	Option D is correct. At high temperature, gas molecules possess high kinetic energy that easily overcomes weak intermolecular attractions. At low pressure, gas molecules are spaced very far apart in a large volume, rendering the physical volume of the molecules completely negligible compared to the container volume. Under these conditions, the two main sources of non-ideality vanish.	[1]
3	A	Option A is correct. At high pressure, molecules are compressed into a small space. The volume occupied by the gas molecules themselves becomes a significant fraction of the total container volume, meaning the free volume available for molecules to move in is less than the container volume ($pV > nRT$, compressibility factor $Z > 1$).	[1]
4	C	Option C is correct. At low temperatures, gas molecules have low average kinetic energy. As molecules pass close to one another, intermolecular attractive forces (London dispersion or dipole-dipole) pull them together. This attraction reduces the momentum and force with which molecules strike the container walls, resulting in an observed pressure lower than predicted by the ideal gas equation ($pV < nRT$, $Z < 1$).	[1]
5	A	Option A is correct. Helium has only 2 electrons per atom, an exceptionally small atomic volume, and extremely weak London dispersion forces. NH_3 and SO_2 possess permanent dipole moments and form hydrogen bonds or strong dipole attractions, while CO_2 has 22 electrons and significant dispersion forces. Helium therefore exhibits the least intermolecular attraction and smallest particle volume, behaving most ideally.	[1]
6	D	Option D is correct. At moderate pressures, intermolecular attractive forces dominate over particle volume effects. Strong intermolecular forces (dipole-dipole and hydrogen bonding in NH_3) draw molecules toward each other, reducing the frequency and force of wall collisions and pulling the molecules closer together, so that pV is less than nRT ($Z < 1$).	[1]
7	B	Option B is correct. $M_r(O_2) = 32.0$. Moles of $O_2 = 0.320 \text{ g} / 32.0 \text{ g mol}^{-1} = 0.0100 \text{ mol}$. Volume at r.t.p. = moles $\times$ molar volume = $0.0100 \text{ mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 0.240 \text{ dm}^3 = 240 \text{ cm}^3$.	[1]
8	C	Option C is correct. Using $pV = (m/M_r)RT \rightarrow M_r = mRT / (pV)$. Convert to SI units: $m = 0.284 \text{ g}$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; $T = 373 \text{ K}$; $p = 100 \text{ kPa} = 100 \times 10^3 \text{ Pa}$; $V = 125 \text{ cm}^3 = 125 \times 10^{-6} \text{ m}^3$. $M_r = (0.284 \times 8.31 \times 373) / (100 \times 10^3 \times 125 \times 10^{-6}) = 880.3 / 12.5 = 70.4 \rightarrow 70.8$.	[1]
9	B	Option B is correct. Density $\rho = \text{mass} / \text{volume} = (p \times M_r) / (RT)$. Using SI units: $p = 101 \times 10^3 \text{ Pa}$; $M_r = 0.0641 \text{ kg mol}^{-1}$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; $T = 298 \text{ K}$. $\rho = (101000 \times 0.0641) / (8.31 \times 298) = 6474.1 / 2476.38 = 2.61 \text{ kg m}^{-3} = 2.61 \text{ g dm}^{-3}$.	[1]
10	A	Option A is correct. Total moles = $0.20 + 0.30 + 0.50 = 1.00 \text{ mol}$. Mole fraction of $O_2 = x(O_2) = 0.30 / 1.00 = 0.30$. By Dalton's law of partial pressures: $p(O_2) = x(O_2) \times P_{\text{total}} = 0.30 \times 200 \text{ kPa} = 60 \text{ kPa}$.	[1]
11	D	Option D is correct. Avogadro's law states that equal volumes of ideal gases at the same temperature and pressure contain identical numbers of molecules (or moles of molecules), regardless of the chemical identity or molar mass of the gas.	[1]
12	C	Option C is correct. According to Graham's law, the rate of diffusion or effusion of a gas is inversely proportional to the square root of its molar mass: Rate proportional to $1 / \sqrt{M_r}$. Since H_2 has the lowest molar mass ($M_r = 2.0$), its molecules possess the highest average root-mean-square velocity at a given temperature, diffusing faster than any other gas.	[1]
13	A	Option A is correct. From the ideal gas law: $(P_1 \times V_1) / T_1 = (P_2 \times V_2) / T_2$. Substituting $V_2 = 0.5 V_1$ and $T_2 = 2 T_1$ gives: $(P_1 \times V_1) / T_1 = (P_2 \times 0.5 V_1) / (2 T_1) \rightarrow P_1 = P_2 \times (0.5 / 2) = P_2 \times 0.25 \rightarrow P_2 = 4 P_1$.	[1]
14	C	Option C is correct. Deviation from ideal behavior increases as intermolecular forces become stronger and molecular volume increases. Helium has only 2 electrons and negligible dispersion forces (least deviation). Nitrogen has 14 electrons and non-polar covalent bonds with weak dispersion forces. Carbon dioxide has 22 electrons with stronger dispersion forces. Ammonia has polar bonds and forms strong intermolecular hydrogen bonds, showing the greatest deviation.	[1]
15	B	Option B is correct. As temperature increases, the average kinetic energy of gas molecules increases. The distribution of molecular speeds broadens, shifting the most probable speed (peak) to the right. Because the total number of molecules is constant (area under the curve must remain constant = 1), the peak height must decrease.	[1]
16	D	Option D is correct. Gas collected over water is saturated with water vapor. According to Dalton's law: $P_{\text{total}} = p(H_2) + p(H_2O) \rightarrow p(H_2) = P_{\text{total}} - p(H_2O) = 102.0 - 2.3 = 99.7 \text{ kPa}$.	[1]
17	C	Option C is correct. $pV = nRT \rightarrow n = pV / (RT)$. Convert units: $p = 2.00 \times 10^5 \text{ Pa}$; $V = 5.00 \text{ dm}^3 = 5.00 \times 10^{-3} \text{ m}^3$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; $T = 300 \text{ K}$. $n = (2.00 \times 10^5 \times 5.00 \times 10^{-3}) / (8.31 \times 300) = 1000 / 2493 = 0.4011 \text{ mol}$. Mass = $n \times A_r = 0.4011 \text{ mol} \times 20.2 \text{ g mol}^{-1} = 8.10 \text{ g}$.	[1]
18	A	Option A is correct. According to kinetic theory, the average translational kinetic energy of gas particles depends SOLELY on thermodynamic temperature ($KE_{\text{avg}} = 3/2 kT$). Since both gases are at the same temperature T , their average kinetic energies are identical. The lighter helium atoms compensate for their smaller mass by moving at much higher speeds.	[1]
19	D	Option D is correct. Ratio of speeds: $v(\text{He}) / v(\text{CH}_4) = \sqrt{M_r(\text{CH}_4) / A_r(\text{He})} = \sqrt{16.0 / 4.0} = \sqrt{4} = 2.0$. Therefore, helium atoms move at exactly twice the average speed of methane molecules at the same temperature.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
20	B	Option B is correct. The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$ is defined in base SI units ($1 \text{ J} = 1 \text{ N m} = 1 \text{ Pa m}^3$). Therefore, pressure MUST be in Pascals (Pa), volume in cubic metres (m^3), and temperature in Kelvin (K). Note that $1 \text{ m}^3 = 1000 \text{ dm}^3 = 1,000,000 \text{ cm}^3$.	[1]
21	B	Option B is correct. By Charles's Law ($V/T = \text{constant at constant } p$): $V_1/T_1 = V_2/T_2 \rightarrow T_2 = T_1 \times (V_2/V_1) = 300 \text{ K} \times (400 \text{ cm}^3 / 600 \text{ cm}^3) = 300 \times (2/3) = 200 \text{ K}$.	[1]
22	D	Option D is correct. In real gases, attractive intermolecular forces reduce the pressure exerted on the walls; the term $a(n/V)^2$ adds the lost pressure back. Real gas molecules have a finite volume (incompressible cores); the term nb subtracts the excluded volume occupied by the molecules themselves from the container volume.	[1]
23	A	Option A is correct. Liquefaction requires intermolecular attractive forces (such as London forces or hydrogen bonds) to overcome thermal kinetic energy and bind molecules into a condensed liquid phase. Because ideal gas particles have zero intermolecular attractions by definition, they can never condense into a liquid.	[1]
24	C	Option C is correct. The critical temperature of a substance is the temperature above which its vapor phase cannot be liquefied by any increase in pressure, because the thermal kinetic energy of the molecules exceeds the maximum possible intermolecular attractive bonding energy.	[1]
25	A	Option A is correct. Boyle's law states that $pV = \text{constant } (k)$ at constant temperature. Rearranging gives $p = k \times (1/V)$. Therefore, plotting p on the y-axis against $(1/V)$ on the x-axis yields a straight line with gradient k that passes directly through the origin (0,0).	[1]
26	D	Option D is correct. Moles of $\text{N}_2 = \text{Volume} / \text{Molar volume} = 0.120 \text{ dm}^3 / 24.0 \text{ dm}^3 \text{ mol}^{-1} = 0.00500 \text{ mol}$. Number of molecules = moles $\times L = 0.00500 \text{ mol} \times 6.02 \times 10^{23} \text{ molecules mol}^{-1} = 3.01 \times 10^{21}$ molecules.	[1]
27	B	Option B is correct. Using Boyle's law $p_1V_1 = p_2V_2$: Initial state: $p_1 = 300 \text{ kPa}$, $V_1 = 1.0 \text{ dm}^3$. Final state: total volume $V_2 = 1.0 + 2.0 = 3.0 \text{ dm}^3$. $p_2 = (p_1 \times V_1) / V_2 = (300 \text{ kPa} \times 1.0 \text{ dm}^3) / 3.0 \text{ dm}^3 = 100 \text{ kPa}$.	[1]
28	C	Option C is correct. Effusion time t is inversely proportional to effusion rate, so t is proportional to $\sqrt{M_r}$. $t(\text{CH}_4) / t(\text{O}_2) = \sqrt{M_r(\text{CH}_4) / M_r(\text{O}_2)} = \sqrt{16.0 / 32.0} = \sqrt{0.5} = 0.7071$. $t(\text{CH}_4) = 50 \text{ s} \times 0.7071 = 35.4 \text{ s}$.	[1]
29	B	Option B is correct. Initial partial pressures: $p(\text{NO}) = 100 \text{ kPa}$, $p(\text{O}_2) = 100 \text{ kPa}$. Reaction ratio: 2 NO reacts with 1 O_2 . NO is the limiting reagent: 100 kPa of NO reacts with 50 kPa of O_2 to form 100 kPa of NO_2 . Remaining unreacted $\text{O}_2 = 100 - 50 = 50 \text{ kPa}$. Total final pressure = $p(\text{NO}_2) + \text{unreacted } p(\text{O}_2) = 100 + 50 = 150 \text{ kPa}$.	[1]
30	A	Option A is correct. When pressure decreases at constant temperature, gas volume increases and the number of molecules per unit volume (number density) decreases. With molecules spaced much further apart on average, a molecule travels a longer distance before colliding with another molecule, increasing the mean free path.	[1]
31	D	Option D is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
32	C	Option C is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
33	A	Option A is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
34	C	Option C is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
35	B	Option B is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
36	D	Option D is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
37	C	Option C is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
38	A	Option A is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
39	D	Option D is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
40	B	Option B is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
41	B	Option B is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
42	D	Option D is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
43	A	Option A is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
44	C	Option C is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
45	A	Option A is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
46	D	Option D is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
47	B	Option B is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
48	C	Option C is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
49	B	Option B is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
50	A	Option A is correct. Under isobaric (constant pressure) conditions for a fixed quantity of ideal gas, Charles's Law dictates that volume is directly proportional to thermodynamic temperature in Kelvin: $V_1/T_1 = V_2/T_2$.	[1]
51	D	Option D is correct. Silicon(IV) oxide (quartz) forms a giant covalent macromolecular network where each silicon is tetrahedrally bonded to 4 oxygen atoms. Iodine forms a simple molecular lattice held by weak London dispersion forces. Magnesium oxide is a giant ionic lattice. Buckminsterfullerene (C ₆₀) is a simple molecular solid.	[1]
52	C	Option C is correct. In diamond, each carbon atom forms 4 strong localized sigma covalent bonds to four neighbouring carbon atoms directed towards the corners of a regular tetrahedron (sp ³ hybridization, 109.5° bond angle). This continuous 3D macromolecular network requires immense energy to disrupt (very high melting point > 3550 °C) and has no mobile delocalised electrons, behaving as an electrical insulator.	[1]
53	A	Option A is correct. In graphite, each carbon atom forms three covalent sigma bonds to three neighbouring carbons in a flat hexagonal planar layer (sp ² hybridization, 120° bond angles). The remaining unhybridized 2p electron on each carbon atom overlaps sideways with adjacent 2p orbitals to form a continuous delocalised pi electron system extending across the entire sheet. These mobile electrons move freely parallel to the layers under an applied potential difference.	[1]
54	C	Option C is correct. While the covalent C-C bonds within each 2D sheet are exceptionally strong (348 kJ/mol), adjacent parallel sheets are separated by 335 pm and held together only by weak intermolecular London dispersion forces. Very little shear force is needed to overcome these weak forces and cause the sheets to slide over one another, imparting softness and lubricating qualities.	[1]
55	B	Option B is correct. Graphene is an isolated single atomic layer of graphite. It consists of sp ² hybridized carbon atoms in a 2D hexagonal honeycomb lattice. The in-plane sigma bonds are among the strongest chemical bonds known, giving graphene immense tensile strength (~130 GPa). Its delocalised pi cloud allows electrons to travel with near-zero effective mass (ballistic transport), exhibiting electrical conductivity superior to copper.	[1]
56	D	Option D is correct. C ₆₀ consists of discrete, hollow spherical molecules resembling a soccer ball (truncated icosahedron: 20 hexagonal rings fused with 12 pentagonal rings). Because it is a simple molecular substance, crystalline C ₆₀ is held together only by weak London dispersion forces between cages. It sublimates at ~600 °C (far lower than diamond/graphite) and dissolves in non-polar organic solvents (e.g. toluene) to form a characteristic magenta solution.	[1]
57	C	Option C is correct. SiO ₂ does not contain discrete molecules. It is a giant covalent network where each Si atom is bonded to 4 oxygen atoms at the vertices of a tetrahedron (109.5° angles), and each oxygen atom acts as a bent bridge connected to 2 silicon atoms (~144° angle). The stoichiometric ratio of Si to O throughout the continuous lattice is 1:2.	[1]
58	A	Option A is correct. Carbon has a smaller atomic radius (77 pm) than silicon (118 pm). In diamond, the C-C covalent bonds are very short (154 pm) and have a high bond enthalpy (348 kJ/mol). The Si-O bonds in SiO ₂ are longer (161 pm) and the Si-O-Si bridging angles allow more flexible lattice deformation under thermal stress, resulting in melting at ~1710 °C.	[1]
59	D	Option D is correct. Solid iodine has a face-centred orthorhombic simple molecular lattice. Inside each I ₂ molecule, two iodine atoms are held by a strong single covalent bond (bond enthalpy 151 kJ/mol). However, the molecules in the lattice are held together only by weak intermolecular London dispersion forces. Gentle heating supplies sufficient kinetic energy to overcome these weak dispersion forces without cleaving the covalent bonds, causing sublime vaporization.	[1]
60	B	Option B is correct. Dynamic equilibrium in a closed system occurs when two opposing processes take place at exactly equal rates: the rate of evaporation (liquid → vapor) equals the rate of condensation (vapor → liquid). At this point, macroscopic properties (such as vapor pressure, liquid volume, and vapor density) remain strictly constant over time.	[1]
61	B	Option B is correct. A liquid boils when its saturated vapor pressure becomes equal to the external ambient pressure. At this temperature, vapor can form bubbles within the interior bulk of the liquid (not merely at the surface), causing rapid boiling throughout the fluid.	[1]
62	D	Option D is correct. Boiling occurs when saturated vapor pressure equals external pressure. Since the external pressure on Mount Everest is only ~33 kPa, water needs to be heated only to ~70 °C for its vapor pressure to reach 33 kPa, whereupon boiling commences.	[1]
63	A	Option A is correct. W is a giant ionic lattice (NaCl-like, high mp, conducts only when molten). X is a giant covalent macromolecular solid (SiO ₂ -like, very high mp, non-conductor, hard). Y is a simple molecular substance (Br ₂ -like, low mp, non-conductor). Z is a metal (Cu-like, high mp, conducts in solid state).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
64	C	Option C is correct. During melting (solid → liquid), the rigid lattice is partially broken; only ~15% of the hydrogen bonds are disrupted, leaving liquid water with extensive hydrogen bonding and molecules in close contact. During boiling (liquid → vapor), molecules must be completely separated to infinite distance against all remaining hydrogen bonds and London forces, requiring vastly more energy.	[1]
65	A	Option A is correct. In solids and liquids, strong chemical bonds or intermolecular forces tightly constrain atomic positions, so thermal expansion is relatively small (~0.01% to 0.1%). In gases, particles are free from constraining attractions, and volume expands proportionally to thermodynamic temperature (Charles's law, ~7% expansion for a 20 K increase at room temperature).	[1]
66	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
67	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
68	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
69	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
70	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
71	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
72	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
73	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
74	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
75	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
76	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
77	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
78	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
79	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
80	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
81	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
82	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
83	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
84	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
85	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
86	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
87	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
88	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
89	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
90	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
91	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
92	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
93	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
94	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
95	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
96	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
97	C	Option C is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
98	A	Option A is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
99	D	Option D is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
100	B	Option B is correct. Giant covalent substances (such as diamond, silicon, and silicon dioxide) consist of continuous arrays of atoms linked by strong covalent bonds extending through the entire crystal. Melting requires breaking thousands of these high-energy covalent bonds, leading to exceptionally high melting points and chemical inertness.	[1]
101	C	Option C is correct. Real gases deviate from ideality when the two core assumptions of the kinetic-molecular theory break down: (1) At high pressure, molecules are packed close together so the physical volume of the molecules is no longer negligible compared to the container volume. (2) At low temperature, molecules move slowly so intermolecular attractions become significant, reducing wall collision force. Hence deviation is greatest at high pressure and low temperature.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
102	A	Option A is correct. Helium atoms possess the smallest atomic radius, only 2 electrons, and no permanent dipole moment, resulting in negligible London dispersion forces and the smallest excluded volume of any gas. HCl, NH ₃ , and CO ₂ all have much larger electron counts and strong dipole or dispersion attractions, deviating significantly more from ideality.	[1]
103	D	Option D is correct. Use $pV = (m/M_r)RT \rightarrow M_r = mRT / (pV)$. Convert all variables into strict SI units: $m = 0.480 \text{ g}$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; $T = 373 \text{ K}$; $p = 101 \text{ kPa} = 101 \times 10^3 \text{ Pa}$; $V = 125 \text{ cm}^3 = 125 \times 10^{-6} \text{ m}^3$. $M_r = (0.480 \times 8.31 \times 373) / (101 \times 10^3 \times 125 \times 10^{-6}) = 1488.5 / 12.625 = 117.9 \rightarrow 118$.	[1]
104	B	Option B is correct. Diamond is a giant covalent macromolecular network held by strong C–C covalent bonds throughout (mp > 3550 °C). MgO is a giant ionic lattice with +2 and –2 ions (mp 2852 °C). NaCl is a giant ionic lattice with +1 and –1 ions (mp 801 °C). Ice is a simple molecular solid held by intermolecular hydrogen bonds that melt at 0 °C.	[1]
105	A	Option A is correct. In graphite, carbon atoms are sp ² hybridized, each bonded to 3 other carbons in planar hexagonal layers. The fourth valence electron occupies an unhybridized 2p orbital that overlaps to form a continuous delocalised pi conduction band across the layers. In diamond, each carbon is sp ³ hybridized and bonded tetrahedrally to 4 carbons; all valence electrons are locked in localized sigma bonds, leaving no mobile charge carriers.	[1]
106	C	Option C is correct. In graphite, the strong covalent bonds exist only within the 2D hexagonal sheets. Between adjacent sheets, there are only weak intermolecular London dispersion forces with an interlayer separation of 335 pm. These weak forces permit the layers to slide smoothly over one another when mechanical stress is applied, making graphite an exceptional dry lubricant.	[1]
107	B	Option B is correct. In the giant covalent lattice of SiO ₂ (quartz), each silicon atom is tetrahedrally bonded to 4 oxygen atoms (coordination number of Si = 4). Each oxygen atom acts as a bridge bonded to 2 silicon atoms (coordination number of O = 2). This 4:2 coordination corresponds precisely to the 1:2 empirical formula SiO ₂ .	[1]
108	D	Option D is correct. Total moles of gas = 0.40 + 0.60 = 1.00 mol. Mole fraction of neon, $x(\text{Ne}) = 0.40 / 1.00 = 0.40$. By Dalton's law of partial pressures: $p(\text{Ne}) = x(\text{Ne}) \times P_{\text{total}} = 0.40 \times 500 \text{ kPa} = 200 \text{ kPa}$.	[1]
109	A	Option A is correct. A liquid boils when its saturated vapor pressure equals the external ambient pressure. At high altitudes, atmospheric pressure is substantially lower than standard sea-level pressure (101 kPa). Consequently, water's vapor pressure matches the surrounding air pressure at a lower temperature (e.g. 85 °C), causing boiling to occur earlier.	[1]
110	C	Option C is correct. Diamond is a giant covalent macromolecular lattice: melting or dissolving it requires cleaving immense numbers of strong localized covalent C–C bonds (348 kJ/mol), making it insoluble and giving it an enormous melting point (>3550 °C). In contrast, C ₆₀ consists of discrete, individual molecular cages. The cages are held together in the crystal only by weak intermolecular London dispersion forces, which are readily overcome by gentle heating (~600 °C) or by solvent interactions with non-polar solvents like toluene.	[1]